Antoni Pietraszewski

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Expected graduation: Feb 2028 | **Available:** 1 Jul 2027, full-time, 13–14 weeks, on-site Kraków, Poland

Education

AGH University of Krakow Kraków, Poland

B.Eng. in ICT Engineering (Teleinformatyka), WIET Faculty; 5th of 7 semesters *Oct 2024 – Feb 2028 (expected)*

- **Relevant coursework:** Algorithms & Data Structures, OOP, Databases, Computer Architecture & Operating Systems, IP Networks, Machine Learning*, DevOps Methodology*, Large-Scale Data Distribution Systems* (*ongoing)
- **Cisco CCNA:** courses 1–2 completed (2026), course 3 (Enterprise Networking) in progress

Experience

Relu BV Jun 2026 – Sep 2026

Software Engineer Intern, ML for 3D dental imaging, on-site *Leuven, Belgium*

- Retrained a production 3D segmentation model used by paying customers, iterating on the data pipeline and architecture, training on cloud GPUs and comparing each version with the existing baseline
- Found why one architecture variant was not learning: the target depended on the whole input, but patch-based training only sees part of it. Reframed the target so it could be learned locally, and the model converged
- Built a landmark-prediction module on voxelised 3D data, including its data pipeline, evaluation scripts and integration
- Fixed infrastructure bugs such as stale cached data being silently reused, missing rerun options and hardcoded paths
- Worked in a production monorepo (PyTorch, ONNX Runtime, Docker); code reviews and pair programming in English

Client Work

WhatsApp Shift Tracker | *Python, FastAPI, PostgreSQL, SQLAlchemy, HTMX, Docker Compose* Sep 2026 – present

- Built a time-tracking system for a Belgian cleaning company (about 40 staff); employees clock in and out by sending START/STOP on WhatsApp, with no new app to learn
- Admin panel with timelines, corrections and Excel reports; every change is audited, and an append-only event log enforced by PostgreSQL triggers meets labour-law rules (5-year retention)
- Runs on my VPS as one Docker image with separate processes for the webhook, panel, scheduler and migrations; 2FA, CSRF and CSP protection, 213 tests; in testing, full rollout Nov 2026

Projects

Dental Aligner Generator | *Python, NumPy, SciPy, trimesh, PySide6* Jan 2026 – Jun 2026

- The first C++17 version offset vertices along their normals, which made the mesh intersect itself in concave areas; rewrote it in Python around a signed distance field (voxelize, EDT, marching cubes), which cannot self-intersect
- Built interactive gingival-margin selection with Dijkstra's shortest path over mesh edges and curvature snapping
- Automated print supports; a parameter sweep cut raft resin about 40% with no loss of adhesion (Form 4B)

RocketSim (in progress) | *Python, PyTorch, MuJoCo, PyBullet, Gymnasium, Stable-Baselines3* 2026 – present

- Building a 6-DoF rocket landing simulator (variable mass, thrust-vector control) with swappable MuJoCo/Bullet backends
- Training a neural landing controller with PPO, randomizing wind, sensor noise, actuator delay and mass
- Designing a parallel benchmark against PID and MPC controllers in unseen conditions and across physics engines

Satellite Simulator | *C++17, FreeRTOS, Python, FastAPI* 2025

- Wrote a FreeRTOS application with 6 concurrent tasks at different priorities, per-recipient message queues and a semaphore-based watchdog, simulating the orbit and an optical link
- Added a Python/FastAPI bridge that reads C++ telemetry over TCP and streams it over REST and WebSocket

Nexora (learning project) | *Java 21, Spring Boot, PostgreSQL, React* Feb 2026 – May 2026

- Built an order system between dental clinics and labs, covering orders, STL-based production queues and invoicing
- Spring Boot backend with 15 REST controllers, 22 JPA entities, JWT role-based access and WebSocket updates

Technical Skills

Languages: Python, C++17, Java, TypeScript, SQL
ML: PyTorch, NumPy, SciPy, ONNX Runtime, RL (Stable-Baselines3), 3D/voxel data, cloud GPU training
Systems & Infra: Linux, Docker, Git, PostgreSQL, FreeRTOS, VPS deployment, VLANs, routing & switching
Backend/Web: FastAPI, SQLAlchemy, Spring Boot, HTMX, React, pytest **Tools:** Claude Code, Codex (used daily)

Languages & Interests

Languages: Polish (native), English (B2) **Interests:** Triathlon (completed a full-distance race), tennis, padel, adventure and strategy board games, flight simulators